

SOH Illusion: Misunderstandings of EV Battery State of Health and Methods to Promote Understanding

Kylie R. Lin
Interactive Computing
Georgia Institute of Technology
Atlanta, Georgia, USA
klin368@gatech.edu

Joey Li
Toyota Research Institute
Los Altos, California, USA
joey.li@tri.global

Jehan Sparks
Toyota Research Institute
Los Altos, California, USA
jehan.sparks@tri.global

Alexandre L. S. Filipowicz
Toyota Research Institute
Los Altos, California, USA
alex.filipowicz@tri.global

David A. Shamma
Toyota Research Institute
Los Altos, California, USA
aymans@acm.org

Laura A. Libby
Toyota Research Institute
Los Altos, California, USA
laura.libby@tri.global

Abstract

Legislation in the USA will soon require that electric vehicles (EVs) display battery degradation in the instrument cluster as “state of health” (SOH), the percentage of the battery’s original capacity. However, the extent to which consumers understand SOH degradation patterns is not known. In an initial study with vehicle owners, we find preliminary evidence for a ‘SOH illusion,’ wherein people expect linear rates of EV battery degradation over time even though batteries degrade non-linearly. Additionally, a third of participants incorrectly confused SOH with a battery’s remaining usable life, demonstrating some misunderstanding of SOH among vehicle owners. In a follow-up study we find that framing SOH information with different chart types and legends reduces linear degradation assumptions and aligns people’s expectations. We discuss implications for the design of SOH representations in user interfaces that vehicle UI designers could employ to promote better EV battery understanding.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Information visualization**; • **Applied computing** → **Law, social and behavioral sciences**.

Keywords

Battery State-of-Health, Battery Degradation, Design Interventions, Electric Vehicles, Behavioral Science

ACM Reference Format:

Kylie R. Lin, Joey Li, Jehan Sparks, Alexandre L. S. Filipowicz, David A. Shamma, and Laura A. Libby. 2025. SOH Illusion: Misunderstandings of EV Battery State of Health and Methods to Promote Understanding. In *17th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '25)*, September 21–25, 2025, Brisbane, QLD, Australia. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3744333.3747828>



This work is licensed under a Creative Commons Attribution 4.0 International License. *AutomotiveUI '25, Brisbane, QLD, Australia*
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2013-0/25/09
<https://doi.org/10.1145/3744333.3747828>

1 Introduction

Electric vehicle (EV) batteries degrade with time and use. Owners of EVs are aware of this potential degradation and keen to behave in ways that reduce wear on their EV’s battery [18]. However, many current EV models have no direct method of monitoring their vehicle’s battery health.

New California legislation aims to help consumers know more about the condition of an EV’s battery. Title 13 §1962.5 of the California Code of Regulations (CCR) requires manufacturers to display an EV lithium ion battery’s state of health (SOH) – the remaining proportion of a battery’s usable capacity from its original capacity [14] – on in-vehicle displays by model year ’26 [32]. A high number of EVs in the U.S. are sold in California (37% of new U.S.-based EV registrations in 2022 [1]). Therefore, SOH information will soon become accessible to a large market of new EV owners.

A challenge with the large-scale release of SOH information is that the general public’s understanding of SOH is not well-researched. There are at least two potential challenges facing consumers’ understanding of SOH. First, new legislation mandates that SOH be displayed as a percentage. However, it is unclear how well people understand the concept of SOH and whether displaying SOH as a percentage could potentially be confused with other more commonly reported battery information, such as the battery’s state of charge (SOC; the charge remaining in a battery).

Second, modern EVs primarily use Lithium-ion (Li-ion) batteries that degrade *non-linearly* over time. SOH declines sharply in the battery’s early stage of life, followed by a period of steady, linear decline, which is then followed by a precipitous drop into the end of the battery’s usable life [3, 22, 40, 41]. However, cognitive psychology research consistently finds that people think *linearly* about trends [23, 27, 46]. It is therefore unclear whether people’s tendency to expect linear trends will create misperceptions about SOH, as has been found for other vehicle-based non-linear trends [28]. For example, if vehicle owners expect batteries to degrade linearly with time, they may worry excessively about normal sharp initial decreases in battery SOH or mistakenly believe that their vehicle has plenty of life when it is nearing the end of its usable life.

In this work, we investigate people’s understanding of SOH and SOH degradation trends over a battery’s lifetime. We first contribute an online survey study that examines how well people understand the term ‘state of health’ and investigates people’s expectations of

battery degradation across the typical Li-ion battery life cycle. We further contribute a second study that assesses whether using charts and legends to frame SOH information aligns people's expectations more closely to prototypical SOH patterns. Our work results in (1) insights into how vehicle owners understand SOH across the life cycle and (2) implications for the design of SOH representations that promote EV battery understanding in consumers.

2 Related Work

Although perceptions of battery health has been a topic in mobile phone research [15, 38], less is known about people's understanding of vehicle batteries. To investigate this research area, we focus on (1) People's understanding of SOH versus related concepts such as SOC, (2) known Li-ion battery degradation patterns that capture SOH decline related to time and use, (3) cognitive biases known to inhibit people's ability to process information, and (4) framing effects that impact people's understanding of different trends.

2.1 Understandings of EV Battery SOH

A battery's SOH corresponds to the percentage of its remaining usable capacity compared to its original capacity [14]. Although battery SOH is an important component of modern EVs, there is little prior work on people's understanding of the metric. Filipowicz et al. found that drivers place great importance on battery health, which could influence how they choose to charge their vehicles [18]. However, some work suggests confusion among the general public about SOH versus other battery-related concepts such as "state of charge" (the ratio of a battery's current amount of charge to the maximum charge it can store [8, 35]). Wang et al. [49] found that displaying SOH in-vehicle reduced people's "range anxiety" (the worry some drivers feel that they will run out of charge in an EV). This suggests that although drivers worry about SOH, they may also confuse displays of SOH, particularly as a percentage, with other battery metrics such as SOC. Due to the limited work in this area, **Study 1** of our investigation determines whether people understand SOH or confuse it with other concepts such as SOC.

2.2 Lithium-ion Battery Degradation

Li-ion batteries are the most common EV battery type [9, 48] and can exhibit a variety of degradation patterns as the battery ages through normal use [2]. However, most EV batteries display non-linear aging trajectories [3, 40, 41], with three main stages of the battery life cycle [22] (Fig. 1A). In the first stage, battery SOH sees a rapid initial decline. In the second stage, SOH enters a long period of steady, moderate decline. In the third stage, SOH drops precipitously into the end of the battery's usable life (e.g., from 70% to below 50% in a quick span of time as compared to decline in the second stage). As per the IEEE standard 1188.1996, the point at which a battery should be replaced is when SOH is less than 80% [7, 44], which anticipates the drop-off (often referred to as a battery 'knee' [2]) that typically occurs when SOH is between 70% and 80%. Importantly, this means that SOH values are not equivalent to the percentage of a battery's remaining usable life. For example, 80% SOH does not mean '80% of the battery's usable life is left', but rather that the battery is near the end of its usable life.

In general, the rate of degradation in these three stages of a Li-ion battery's life can vary based on factors such as battery chemistry, operating temperature, and charging profiles (e.g., speed at which the battery is charged or discharged) [20, 37]. However, the graph of the entire aging trajectory consistently resembles a characteristic S-shape sigmoid function, even if the exact shape may vary between batteries. In this article, we refer to these three stages as the early life stage, the middle life stage, and the late life stage.

2.3 Cognitive Biases in Vehicle Understanding

Cognitive biases are aspects of human cognition that inhibit people's ability to accurately interpret information and make judgements [23]. Psychology research has specifically examined biases in the context of driving. Eriksson et al. found support for the time saving bias in driving behaviors, wherein people overestimated the amount of time saved by driving at a higher speed to reach their destination [12]. Svenson et al. also found a braking capacity bias, where people are overly optimistic about their ability to slow down when braking at high speeds [42, 43].

Cognitive biases also influence the way people perceive trends. Larrick and Soll found support for an "MPG illusion," where people expect a vehicle's fuel consumption to decrease linearly as a function of miles-per-gallon (MPG), whereas in reality the benefits of increased MPG diminish curvilinearly (e.g., the fuel savings of increasing MPG from 50 to 55 are smaller than increasing MPG from 20 to 25) [27, 28]. Underlying this phenomenon is extensive cognitive psychology literature showing that people have difficulty understanding non-linear relationships. For example, Kahneman and Tversky introduced the representativeness heuristic and anchoring bias, which can lead people to make simplified judgements that assume linear relationships [46]. Kalish et al. demonstrated that people tend to perceive linear trends when observing data generated by non-linear processes [24]. Although these biases can be mitigated through visualizations and interaction with data (e.g., if people are asked to draw trend lines between data points), without additional context, people default to expecting linear trends [11]. Drawing on this prior work, we hypothesized that people expect SOH to degrade at a linear rate over time. To test this hypothesis, **Study 1** of our investigation measured people's expectations around SOH through typical phases of a Li-ion battery's life cycle.

2.4 Framing Effects

As a solution to the MPG illusion, Larrick and Soll found that flipping the units of measurement to Gallons per Mile helps people reason more accurately about internal combustion engine vehicle efficiency [28]. Their solution demonstrates the power of framing effects, where altering how information is presented results in people making different judgments [36, 45]. Framing effects have also been studied in the context of ride-hailing decisions. Mohanty et al. found that framing ride-share options with negatively valenced wording and information about popularity increased people's likelihood of selecting eco-friendly rides [30].

Beyond framing information in plain text formats, information visualization research has found that the way that data is visually presented impacts how it is processed [19, 26]. Visualizations can enhance numerical understanding of data, emphasize trends, and

evoke emotional responses [31, 52]. Longstanding work by Cleveland and McGill shows that chart types that encode data via position support more accurate data interpretation [10]. More recently, Xiong et al. found that text excerpts, bar charts, line charts, and scatterplots each activate different cognitive reasoning affordances [51]. Similarly, Bearfield et al. found that different data patterns become more or less evident to readers depending on if the data is represented as a bar chart as opposed to a table; moreover, even when viewing the same visualization, different people can reach different conclusions [5, 17]. Beyond data interpretation, work by Pandey et al. found that charts can be effective tools for persuasion [34].

Based on this prior work, **Study 2** of our investigation examines whether framing SOH information via different chart types improves people’s battery degradation expectations. Gauge charts have long been used in the vehicle instrument cluster to communicate information such as driving speed [21]. Thus, we examine the gauge chart and its non-circular counterpart, the bullet chart. Research on the efficacy of gauge and bullet charts focuses on time-sensitive chart-reading tasks and offers conflicting findings – with some suggesting that vertical and horizontal gauge displays are more difficult to read than circular ones [21] and others concluding that horizontal gauges are more efficient and less visually demanding to interpret [50]. Given the lack of work specifically assessing gauge and bullet charts in communicating SOH, a metric that changes gradually over numerous driving sessions, we test both as a starting point for supplementing SOH information in-vehicle.

3 Study 1: People’s Understanding of SOH

Study 1 investigates whether vehicle owners understand the meaning of battery SOH (as opposed to SOC) and their expectations for battery degradation across the three stages of battery life. We conducted a within-subjects study aimed at testing people’s knowledge and expectations about EV Battery degradation. After first being informed that EV batteries will naturally degrade with time, participants answered (1) questions assessing their understanding of SOH and (2) a set of hypothetical text-based scenarios to capture their ability to predict future SOH across early, middle, and late stages of battery life. The main research questions and hypotheses for this study can be seen in Table 1.

3.1 Participants

We recruited 558 US-based participants from Prolific [33] for study 1 (53% women, 44% men, 2% genderqueer or non-binary, <1% agender, and <1% preferring not to answer; median age range = 26–35 years). Participants were at least 18 years of age, fluent in English, and were vehicle owners (88.5% Gas, 7.2% Hybrid, 4.1% Electric, <1% elected not to answer). This study was reviewed and approved by Western IRB and all participants provided informed consent before beginning the task. Participants were paid \$6 for participation.

3.2 SOH Understanding Questions

Participants first saw two questions assessing their initial understanding of SOH. The first was a multiple-choice *definition* question asking “Which option best describes what you think ‘90% state-of-health (SOH)’ means for an electric vehicle battery?”. Possible responses were “The battery can hold 90% of its original capacity”

Study 1

Q. 1.1	<i>To what extent do vehicle owners understand the concept of SOH?</i>
H. 1.1	Vehicle owners confuse SOH with SOC.
Q. 1.2	<i>How do vehicle owners expect SOH to change over time?</i>
H. 1.2	Vehicle owners expect SOH to degrade at a linear rate across early, middle, and late stages of battery life.

Table 1: Research questions and hypotheses for Study 1.

(correct response), “The battery is 90% charged” (referring to SOC), “90% of the battery’s usable life is left”, “9 years of battery life are left”, and “I don’t know”. The second was a *mathematical reasoning* question that asked participants to convert between driving range and SOH: “Imagine that a new electric vehicle battery has a 200-mile range. After two years of use, the battery now shows 90% state-of-health (SOH). What is the vehicle’s current maximum range when fully charged?”. Participants answered using a slider widget ranging from 0 to 200 (the correct answer was 180, or 90% of 200).

3.3 Battery Degradation Scenarios

Participants were presented with a hypothetical scenario in which they were asked to imagine that they had just purchased a used EV (see Fig. 1). This scenario contained two parts:

Timepoint 1: Battery health of the used EV starts out at a certain percentage (we refer to this percentage as ‘X’) in 2023 and degrades to a certain percentage (‘Y’) in 2024. Participants were asked to predict SOH for the following year (2025) using a slider widget ranging from 0 to 100.

Timepoint 2: Participants were told that in 2025, battery health had degraded to a percentage (‘Z’). They then indicated actions they would take upon learning how the battery has degraded. We grouped actions into 3 overarching categories, aiming to capture the practical relevance of SOH changes to vehicle owner behaviors: *Further EV Understanding*, *Modify Personal Behaviors*, and *Contact Dealer/Maker* (Table 2).

Participants viewed this scenario three times in randomized order, with percentages X, Y, and Z substituted for early, middle, and late stages of battery life as described below:

Early life stage: Battery health degrades sharply by 6% between 2023 ($X = 98\%$) to 2024 ($Y = 92\%$) and then by a more modest 2% between 2024 to 2025 ($Z = 90\%$). Participants predicting linear degradation patterns would respond 86% (i.e., a 6% degradation trend into the next year).

Middle life stage: SOH degrades by 2% between 2023 ($X = 86\%$) to 2024 ($Y = 84\%$) and continues at a modest 2% decline between 2024 to 2025 ($Z = 82\%$). Participants predicting linear degradation patterns would respond 82% (i.e., a 2% degradation trend into the next year).

Late life stage: SOH degrades modestly by 2% from 2023 ($X = 72\%$) to 2024 ($X = 70\%$). Between 2024 and 2025, participants are informed that the battery “no longer works to operate the electric vehicle and needs to be replaced” ($Z = \text{end of life}$). Participants predicting linear degradation patterns would respond 70% (i.e., a 2% degradation trend into the next year).

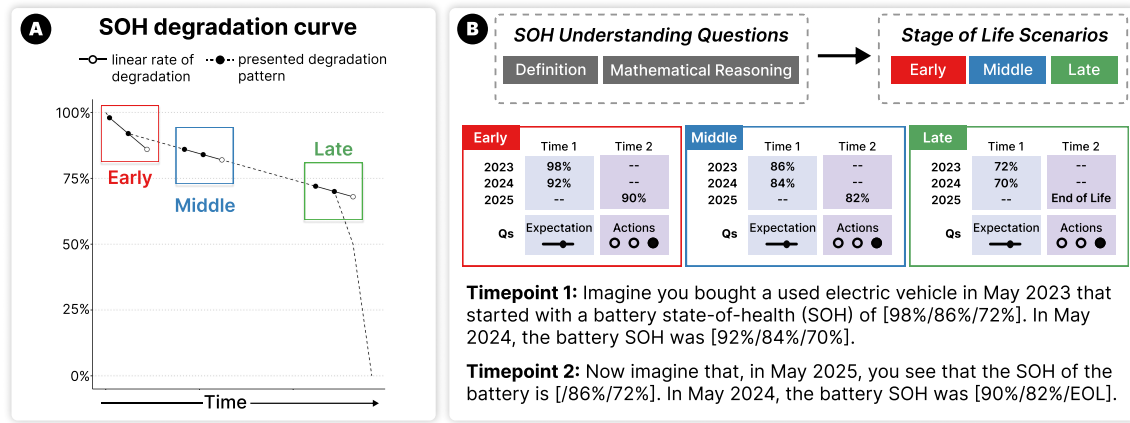


Figure 1: (A) The example battery life trajectory we present to participants in Study 2. (B) The setup for Study 1. We conduct a within-subjects experiment with three repeated measures (early, middle, and late life stages). Participants first answer SOH understanding questions and then see a series of Battery Degradation Scenarios that represent unique battery life stages.

As noted in the introduction, real-world battery degradation trends can vary based on factors such as battery chemistry, operating temperature, and charge/discharge cycles [20, 37]. Therefore, the values we selected are not representative of *all* degradation trends across different EV models. However, we chose values that match the typical sigmoidal Li-ion battery degradation patterns and therefore suited to test reactions to normal degradation trends.

3.4 Analysis

Analyses were conducted in R. To compute the likelihood that people made linear degradation predictions across different battery life stages, we used a logistic mixed effects (LME) model from the `lme4` library in R [4], using life stage as a fixed effect and participants and life stage order as random intercepts. To analyze the likelihood of considering various actions across the battery life cycle, we first examined the percentage of participants who indicated that they would consider actions that fell into each category. Next, we ran 3 LME models to measure effects of life stage. The first model predicted the likelihood of selecting at least one action under *Further EV Understanding*, the second predicted the likelihood of selecting at least one action under *Modify Personal Behaviors*, and the third predicted the likelihood of selecting at least one action under *Contact Dealer/Maker*. All three models used life stage as a fixed effect and participants and life stage as random intercepts. We set the middle life stage scenario as a baseline in each model because it represents steady degradation over multiple years, whereas early and late life stage scenarios each depict non-linear rates of degradation. In the following results section, we present an overview of main findings; for more detailed statistical reporting, please see supplemental materials available in the Open Science Framework ¹.

3.5 Results

3.5.1 Most participants ‘get’ SOH, but some think that it means percentage of remaining usable life. Overall, 61.83% of participants

Further EV Understanding

Understand how battery life is related to driving range.
Understand how battery health impacts vehicle value.
Understand more about why battery life matters.
Learn more about EVs in general.
Learn about ways to improve battery health.

Modify Personal Behaviors

Change my driving behavior.
Change my charging behavior.

Contact Dealer/Maker

Provide feedback about the car.
Submit a battery warranty claim.

Table 2: Answer options for the question: “What actions would you consider taking after learning this information?”

answered correctly when asked what was meant by “90% state-of-health (SOH).” Of the 38.17% who answered incorrectly, most answered “90% of the battery’s usable life is left” (32.80% of total responses), indicating that roughly a third of surveyed vehicle owners think SOH refers to the percentage of usable life in the battery. Only 3.58% of the participants selected “The battery is 90% charged,” the description that described SOC, 1.08% selected “I don’t know,” and less than 1% selected “9 years of battery life are left.” For the *mathematical reasoning* question, 62.90% of participants answered exactly correctly, demonstrating that most participants were able to convert between SOH and driving range. For this question, we note that the use of a slider scale may have led some participants who meant to indicate 180 to enter their answer slightly higher or lower than 180; our analysis provides a strict assessment that still indicates a high degree of understanding of how to convert between SOH and driving range.

3.5.2 Many participants expected EV batteries to degrade at a linear rate across all life stages. A large proportion of participants expected linear battery degradation across all life stage scenarios (46.48% overall), predicting that the rate of degradation from 2024

¹OSF project accessible at: osf.io/78p2f



Figure 2: (A) Distribution of Responses to the prediction question. Colored points indicate predicted answers if people were making linear responses and violin plots represent the density of individual participant SOH predictions for each battery life stage. (B) The likelihood of making linear responses across lifestage scenarios.

to 2025 would be the same as the rate they saw from 2023 to 2024. In the early life stage scenario, median SOH predictions across all participants was 86%, the exact number expected if people were making linear predictions (mean = 84.41, 95%CI = [83.85, 84.97]; Fig. 2A). In the middle life stage, the median SOH predictions 82, also the exact number expected if participants made linear predictions (mean = 79.74, 95%CI = [79.18, 80.29]; Fig. 2A). For the late stage, the median SOH prediction was 68, again indicating linear expectations (mean = 66.69, 95%CI = [66.24, 67.14]; Fig. 2A). Therefore, across all lifestage conditions, median predictions of battery degradation exactly matched what would be expected if participants were expecting linear battery degradation trends. We note that the mean values were lower than all median values because the distribution of participant responses were left skewed across all life stages (Pearson coefficient of skewness: $\text{skew}_{\text{early}} = -4.99$, $\text{skew}_{\text{mid}} = -5.75$, $\text{skew}_{\text{late}} = -3.39$).

Although participants tended to make linear responses across all battery life stages, the likelihood that they would make a linear response varied. Participants were 2.1x less likely to make an exact linear prediction in the early life stage scenario compared to the middle life stage scenario (LME model predicting the likelihood of making an exact linear response: $\beta = 0.72$, $z = 4.78$, $p < 0.001$; Figure 2B) and 2.6x less likely compared to the late life stage scenario ($\beta = 0.94$, $z = 6.16$, $p < 0.001$). Participants were equally likely to make linear predictions in the late and middle life stage scenarios ($p = 0.142$). Importantly, this indicates that although people are less likely to assume that linear degradation trends continue in early life, they expect linear trends to continue in later stages when the battery is most likely to fail.

3.5.3 In general, people want to learn more about EVs but don't want to change their driving or charging behaviors. Across all life stage scenarios, when asked about the actions they would take after learning about battery degradation trends, 78.5% of participant indicated that they would consider an action in the *Further EV Understanding* category (Figure 3). Across all life stages, only 30.8% of participant responses included an action in the *Modify Personal Behaviors* category. Across all life stages, 38.5% of responses included an action

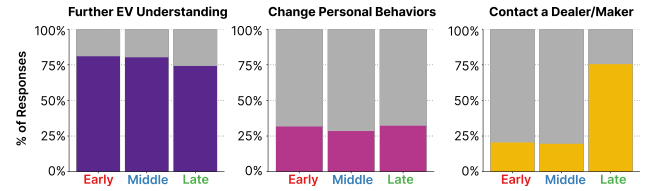


Figure 3: Proportion of Study 1 participants that selected actions under the three established categories.

in the *Contact a Dealer/Maker* category. However, we found that responses for this category differed by life stage (20.4% and 19.5% in the early and middle life stages, but 75.4% in the late life stage). We examine the extent of differences in responses across life stage in the following section.

3.5.4 Unexpected end of life increases negative outcomes. Participants were 34.4x more likely to select actions from the *Contact a Dealer/Maker* category in the late life stage as opposed to in the early life stage ($\beta = 3.53$, $z = 14.57$, $p < 0.001$; Fig. 3) and 37.0x more likely that in the middle life stage ($\beta = 3.61$, $z = 14.69$, $p < 0.001$). Moreover, participants were less 1.7x likely to select actions in the *Further EV Understanding* category in the late life stage as opposed to in the early life stage ($\beta = 0.55$, $z = 3.18$, $p < 0.01$) and 1.6x less likely than in the middle life stage ($\beta = 0.48$, $z = 2.83$, $p < 0.01$). There was no significant difference in participants' likelihood of selecting actions under *Further EV Understanding* ($p = 0.724$) or *Contact a Dealer/Maker* ($p = 0.679$) in the early life stage compared to the middle life stage. Finally, there was no significant difference in participants' likelihood of selecting actions under *Modify Personal Behaviors* between any of the life stages (all $ps > 0.092$).

3.6 Findings

Overall, participants did not confuse SOH with SOC, which does not support hypothesis **H1.1**. However, there was still confusion about SOH. Roughly a third of participants incorrectly thought that SOH refers to the percentage of usable life in a battery. Conversely, we found strong support for hypothesis **H1.2**. The median participant

expectation ratings across all life stages were exactly the values of predicted SOH that constituted a linear prediction, with just under half of participants reporting a number that exactly predicted a linear rate of degradation (Fig. 2). Participants were less likely to make a linear prediction in the early life stage than in the middle or late life stages but equally likely in the middle and late stages.

These findings indicate that vehicle manufacturers and policy makers designing SOH interfaces should expect that a number of vehicle owners will be confused about SOH and how it changes over time. It is therefore important to consider methods of communicating SOH that mitigate misunderstandings before they occur. Our next study explores and evaluates methods to better inform vehicle owners for the possibility of non-linear degradation patterns.

4 Study 2: Frames for SOH Understanding & Taking Action

Results from Study 1 provide insights into people's general understanding of SOH and degradation patterns across the battery life cycle. A substantial portion of vehicle owners misunderstand SOH, conflating it with remaining usable life and incorrectly assume that the rate of battery degradation to be linear in early and late battery life stages. It is well-established that 'framing' can adjust the way information is processed [36, 39, 45], and framing has been found to impact people's transportation and charging decisions [18, 30]. Thus, Study 2 examines methods of framing SOH to improve understanding. Study 2 explores the impact of visualization framing—gauge charts, bullet charts, and both chart types with legends—on people's expectations of future SOH. Our main research questions and hypotheses are described in Table 3.

4.1 Experiment Setup

The experimental setup used in Study 2 was nearly identical to the setup in Study 1, with two key differences. First, participants were divided into five between-subject conditions, each with distinct visualizations used to display battery SOH (Fig. 4). All stimuli contained the title "My Battery Health" and a rating label as a subtitle corresponding to the depicted SOH value. Each label had an associated color: 'Excellent' was dark green, 'Good' was light green, 'Fair' was yellow, 'Poor' was red, and a dark gray color represented the battery being no longer usable. These colors were also used for the SOH value depicted alongside each of the charts. We describe each condition in detail below:

Baseline percent ($n = 397$): participants were exposed to the same text-based experiment as in Study 1, where SOH was displayed as a numeric percentage with no additional visualizations. **Gauge** ($n = 399$): SOH was displayed as part of a gauge chart with colored sections aligning with each of the SOH ranges associated with different battery health rating labels (Fig. 4A). **Bullet** ($n = 401$): SOH was displayed as part of a bullet chart with colored sections aligning with each of the SOH ranges associated with different battery health rating labels (Fig. 4B). **Gauge with legend** ($n = 397$): SOH was displayed as part of the chart as in the gauge condition but with an additional legend listing battery degradation rate for each SOH range as defined by battery health rating label. 'Excellent' SOH was mapped to 'normal or faster degradation,' 'Good' and 'Fair' SOH was mapped to normal degradation, 'Poor' SOH was mapped

Study 2

- | | |
|---------------|--|
| Q. 2.1 | <i>How does framing battery SOH with visualizations and with information about battery degradation affect expectations of future SOH?</i> |
| H. 2.1 | Visualizations and information about battery degradation reduce linear expectations of future SOH and better align expectations to match the typical Li-ion battery degradation curve. |

Table 3: Research question and hypothesis for Study 2.

to 'faster than normal' degradation, and 'End of Life' was mapped to the description 'Battery will no longer function.' (Fig. 4C). **Bullet with legend** ($n = 401$): SOH was displayed as part of a bullet chart with the same battery degradation legend included in the gauge with legend condition (Fig. 4D).

The second key difference from Study 1 was the format of the prediction questions. Rather than a 0-100 slider widget, participants were presented with a multiple-choice selection of possible SOH figures. For the the baseline percent condition, answer options were displayed as plain text (i.e., 90%, 88%, 86%, 84%, and "None—the battery will need to be replaced"). For the visualization-based conditions, images of the data dashboard representing a SOH value for each answer option were provided (for the full suite of possible MCQ answer options, please see supplemental materials).

4.2 Participants

We recruited 1995 US-based participants from Prolific (53% women, 45% men, 1% genderqueer or non-binary, >1% agender, >1% preferring not to answer or to self-describe; median age range = 36–45 years). Participants were at least 18 years of age, fluent in English, and were vehicle owners (86% Gas, 9% Hybrid, and 5% Electric). This study was reviewed and approved by Western IRB and all participants provided informed consent before beginning the task. Participants were paid \$5.67 for participation.

4.3 Analysis

To examine the effects of label condition and life stage on linear degradation predictions, a Type III analysis of variance (ANOVA) test was applied to a LME model using condition and life stage as fixed effects and participants and life stage order as random intercepts. To determine differences between visualization conditions within life stage (e.g., comparing early life stage responses from participants in the baseline percent condition to early life stage responses from participants in the gauge condition), we performed the general linear hypothesis test (GLHT) for post-hoc pairwise comparisons using Tukey's HSD test.

4.4 Results

4.4.1 All visual representations decreased the likelihood of linear degradation predictions. Across all conditions and life stage scenarios, slightly less than half of participant predictions of battery degradation were linear (47.48%). Similar to the results from Study 1, participants in the baseline condition expected linear rates of degradation: the median SOH prediction was 86% in the early life stage scenario (mean = 80.22, 95%CI = [79.40, 81.04]), 82% for the middle life

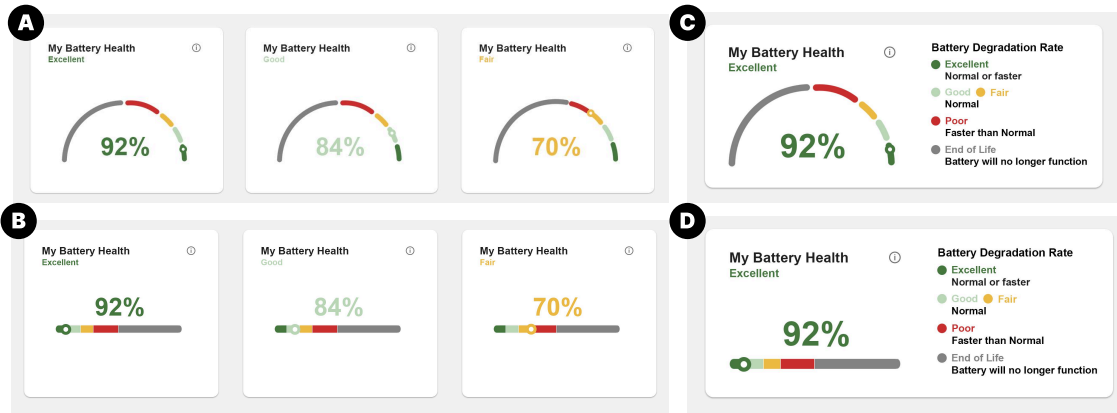


Figure 4: Example stimuli from Study 2. (A) gauge, (B) bullet, (C) gauge with legend, and (D) bullet with legend.

stage scenario (mean = 66.40 (95%CI = [65.64, 67.15]), and 68% for the late life stage scenario (mean = 66.40 (95%CI = [65.64, 67.15]; see Fig. 5A).

However, all visualization conditions substantially reduced people's tendency to make linear predictions of battery degradation across all battery life stages (Fig. 5). We found significant effects of visualization condition ($\chi^2(4) = 180.26, p < 0.001$) and life stage ($\chi^2(2) = 276.33, p < 0.001$) on whether predictions of future battery SOH were linear. Post-hoc analyses revealed that participants in the gauge condition were 3.9x, 3.1x, and 2.0x less likely to make a linear prediction in the early, mid, and late stages, respectively, than in the baseline condition (all β s > 0.69, all z s > 6.73, all p s < .01). Likewise, participants in the bullet condition were 4.3x, 4.0x, and 2.2x less likely to make linear predictions in the early, mid, and late stages, respectively, compared to baseline (all β s > 0.78, all z s > 4.46, all p s < .001).

The same trends held for gauge and bullet visualizations with legends. Participants in the gauge condition with legends were 4.4x, 3.7x, and 2.3x less likely to make a linear prediction in the early, mid, and late stages, respectively, than in the baseline condition (all β s > 0.82, all z s > 4.80, all p s < .001). Likewise, in the bullet condition with legends, participants were 3.7x, 3.9x, and 2.6x less likely to make a linear prediction in the early, mid, and late stages, respectively, than in the baseline condition (all β s > 0.94, all z s > 5.32, all p s < .001). However, the added information provided by legends did not provide any benefit above and beyond the effect already provided by gauge and bullet charts (see Fig. 5C). Taken together these results provide strong support that people's linear assumptions about battery degradation can be mitigated by different visual framings, such as the gauge and bullet charts tested here, and that legend information does not provide any additional benefit.

4.4.2 Visualizations were less effective at reducing linear assumptions in the late battery life stage. Although visualization framing reduced participant's linear assumptions about battery degradation across all battery life stages, their effect was less pronounced in the later life stage. Our earlier ANOVA found a marginal interaction effect between visualization condition and life stage ($\chi^2(8)=15.42, p = 0.052$). This interaction was driven by smaller changes in the

likelihood of making linear assumptions in the late battery life stages than in either the early or mid-life stages (Fig. 5C). Therefore, although visualizations reduced linear assumptions for all stages, they have less of an effect at the late stage.

4.5 Study 2 Findings

The results from Study 2 provide support for hypothesis H2.1. Charts and charts with legends lowered participants' likelihood of making linear predictions, and this across all battery life stage scenarios (Figure 5). However, the information provided by legends did not have any additional effect above and beyond the visual framing provided by gauge and bullet charts. Additionally, although visualizations reduced the likelihood of linear degradation assumptions across all battery life stages, their influence was weaker for late battery stage predictions.

Taken together, these results provide strong support that visualization framing (e.g., gauge and bullet charts) can help mitigate cognitive biases that lead people to expect linear battery degradation and can better align people's expectations with real degradation trends. We finish by discussing the implications of our results for user interface designers developing in-vehicle SOH representations.

5 Discussion

The current article focused on people's conceptions of SOH and battery degradation as it may be presented to consumers in the vehicle instrument cluster. Study 1 provides support for a 'SOH illusion,' demonstrating that people expect SOH to decrease linearly across the battery's usable lifetime, even though real degradation is non-linear. Study 2 explores methods of mitigating people's bias towards linear trends, finding that visual framing of SOH information through gauge and bullet charts help better align people's expectations to realistic degradation patterns. In the following sections we discuss the implication of the results for the design of in-vehicle displays of SOH information.

5.1 Support for a SOH Illusion

Our work provides initial support for a 'SOH illusion' by showing that just under half of vehicle owners predict EV battery SOH

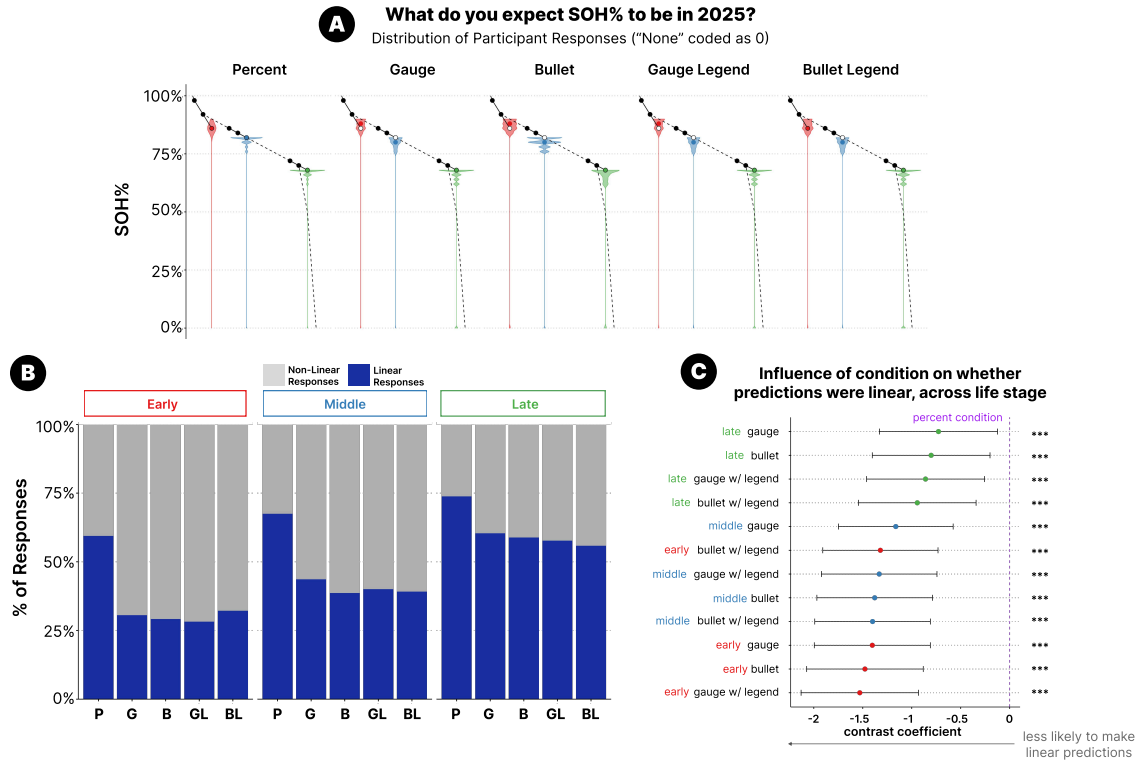


Figure 5: Participant predictions of future SOH across conditions for Study 2. (A) Distributions of responses for all scenarios. (B) The linearity of responses across life stage and condition. (C) The logistic mixed-effects (LME) contrast coefficients comparing the change in likelihood that participants in the non-baseline conditions made a linear prediction compared with the baseline percent condition, across all life stages. Points indicate contrast beta weights and error bars indicate 95% confidence intervals.

to degrade at a linear rate over time. We additionally found that these tendencies are not due to an inability to convert SOH into a usable metric (e.g., mileage). Rather, our results mirror previous research in cognitive science that, without additional framing or support, people default to expecting linear relationships between variables, even if the underlying trends are non-linear [11, 24, 27, 28]. Indeed, our results bear resemblance to the “MPG Illusion” [27, 28], which can cause people to overestimate the fuel savings they might experience by comparing vehicle MPG.

These tendencies can have a similar effect on people’s expectations about battery degradation. People may worry excessively about early steep degradation patterns, expecting that this steep degradation will continue throughout the vehicle life cycle. Moreover, people purchasing an older used EV may erroneously think that a vehicle has plenty of usable life when it’s battery is in fact near its end of life. Indeed, roughly a third of participants indicates that they thought SOH represented the battery’s usable life, suggesting that a high proportion of current vehicle owners may have misguided expectations (Figure 2).

These results highlight that people’s current understanding of SOH is limited, and that methods are needed to better align people’s expectations with real battery degradation. The following section highlights the results of our visualization study and their implications for in-vehicle displays of SOH.

5.2 Fostering State of Health Understanding

California’s upcoming legislation will expose a large proportion of EV owners in the U.S. to SOH information, starting in late 2025. As highlighted in the previous section, understanding of SOH is generally low and people’s expectations do not align with the common Li-ion battery aging trajectories. Using the methods we tested to frame SOH information in Study 2, we articulate the following implications for the design of the electric vehicle instrument cluster.

5.2.1 Use Thoughtful Visualization Designs to Accompany SOH.

All visualization types tested in Study 2 better aligned participant expectations of future SOH degradation trends. We consider design decisions made in the construction of the stimuli for Study 2 that could have driven these observed effects: color, chart type, and displayed text.

Color. The gauge and bullet charts were colored to semantically represent battery health ratings (Excellent & dark green, Fair & yellow, etc.). Semantically resonant colors have been theorized to lessen cognitive load in information processing [29], and colors have been shown to have a strong impact on people’s emotions [47] and their perception of information represented in data dashboards [6]. Thus, it is possible that color affected participant

predictions of future SOH. When viewing the early life stage scenario, participants viewed SOH in the dark green color and predicted rates of degradation that were less linear and more aligned with the true degradation pattern. When viewing SOH in the middle life stage scenario, participants viewed SOH in a light green shade and made less linear and therefore less aligned predictions of future battery SOH. Future work should investigate the impact of chart type and color separately, to distinguish what specific features of visualization design impact battery degradation understanding.

Chart Type. We selected gauge and bullet charts for this work due to their simplicity and their prevalence in existing vehicle instrument cluster designs. However, we acknowledge that there are limitations to this decision. For gauge charts, information visualization research consistently finds that people have difficulty interpreting graphs based on angles and curvature [10, 13]. Moreover, other visualization types can be more effective for communicating change over time, such as line charts and scatter plots [16]. Future work should more thoroughly investigate what visualization types and designs best promote understanding of different battery degradation patterns.

Displayed Text. Our designs contained a variety of additional textual information besides SOH: a title, a battery health rating label as a subtitle, and a legend explicitly describing battery degradation rates across the life cycle. The manipulation of any one of these features may have an effect on the way that SOH is perceived over time. For instance, slanted framings of titles influence the way that people perceive news articles and data visualizations [25]. Future work could examine alternative framings of SOH that manipulate titles in combination with charts.

5.2.2 Set expectations for late life degradation and the battery knee. In the late life stage scenario, participants consistently failed to anticipate the ‘knee’ that often occurs as a part of Li-ion battery degradation. Study 2 further found that although charts reduced linear predictions of future SOH in late life stage scenarios, the effect was weaker compared to early and middle life stage scenarios. Thus, future work should examine additional framings of SOH and design interventions to warn vehicle owners of potential non-linear degradation.

6 Conclusion & Future Work

Motivated by upcoming changes in California’s regulation around SOH displays, we investigated the extent to which vehicle owners understand EV battery SOH and patterns of degradation. We found support for a ‘SOH illusion’ in that a significant portion of our surveyed sample expected SOH to degrade at a linear rate over time. We also investigated the effect of charts and information as visual frames to improve people’s understanding and expectations of real SOH trends. We found that charts and charts with battery degradation legends better align people’s expectations around SOH. Our findings suggest that leveraging visualizations in the vehicle instrument cluster to accompany SOH may be beneficial; however, we acknowledge that design decisions in the creation of our study stimuli could have driven the effects observed in our studies. As such, we recommend that future work examine how variations in

color, chart type, and text (as well as other design factors) affect people’s understanding of SOH.

More generally, our work found a need for greater education surrounding SOH for the general public. Vehicle owners in our study confused SOH with percentage of remaining usable life and consistently expected SOH to degrade linearly from one year to the next. Although we found that visual representations accompanying SOH can affect expectations of future battery degradation, more research is needed into effective interventions that can be deployed both within the vehicle instrument cluster and also beyond. This line of work will help determine how best to represent SOH to consumers in a way that minimizes misconceptions and promotes calibrated expectations of how SOH will change over the course of the EV battery life cycle.

References

- [1] United States Energy Information Administration. 2023. California leads the United States in electric vehicles and charging locations - U.S. Energy Information Administration (EIA). [https://www.eia.gov/todayinenergy/detail.php?id=61082#:~:text=California%20has%20led%20the%20United,Energy%20Data%20System%20\(SEDS\).](https://www.eia.gov/todayinenergy/detail.php?id=61082#:~:text=California%20has%20led%20the%20United,Energy%20Data%20System%20(SEDS).)
- [2] Peter M Attia, Alexander Bills, Ferran Brosa Planella, Philipp Dechent, Goncalo Dos Reis, Matthieu Dubarry, Paul Gasper, Richard Gilchrist, Samuel Greenbank, David Howey, et al. 2022. “Knees” in lithium-ion battery aging trajectories. *Journal of The Electrochemical Society* 169, 6 (2022), 060517.
- [3] Tobias C Bach, Simon F Schuster, Elena Fleder, Jana Müller, Martin J Brand, Henning Lormann, Andreas Jossen, and Gerhard Sextl. 2016. Nonlinear aging of cylindrical lithium-ion cells linked to heterogeneous compression. *Journal of Energy Storage* 5 (2016), 212–223.
- [4] Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. 2014. Fitting linear mixed-effects models using lme4. *arXiv preprint arXiv:1406.5823* (2014).
- [5] Cindy Xiong Bearfield, Lisanne van Weelden, Adam Waytz, and Steven Franconeri. 2024. Same Data, Diverging Perspectives: The Power of Visualizations to Elicit Competing Interpretations. *IEEE Transactions on Visualization and Computer Graphics* (2024).
- [6] Palash Bera. 2016. How colors in business dashboards affect users’ decision making. *Commun. ACM* 59, 4 (2016), 50–57.
- [7] Lluc Canals Casals and Beatriz Amante García. 2017. Second-life batteries on a gas turbine power plant to provide area regulation services. *Batteries* 3, 1 (2017), 10.
- [8] Wen-Yeau Chang. 2013. The state of charge estimating methods for battery: A review. *International Scholarly Research Notices* 2013, 1 (2013), 953792.
- [9] Weidong Chen, Jun Liang, Zhaohua Yang, and Gen Li. 2019. A review of lithium-ion battery for electric vehicle applications and beyond. *Energy Procedia* 158 (2019), 4363–4368.
- [10] William S Cleveland and Robert McGill. 1984. Graphical perception: Theory, experimentation, and application to the development of graphical methods. *Journal of the American statistical association* 79, 387 (1984), 531–554.
- [11] Michael Correll and Jeffrey Heer. 2017. Regression by eye: Estimating trends in bivariate visualizations. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 1387–1396.
- [12] Gabriella Eriksson, Ola Svenson, and Lars Eriksson. 2013. The time-saving bias: Judgements, cognition and perception. *Judgment and decision making* 8, 4 (2013), 492–497.
- [13] Stephanie DH Evergreen. 2019. *Effective data visualization: The right chart for the right data*. SAGE publications.
- [14] Zhang Fan, Xing Zi-xuan, and Wu Ming-hu. 2023. State of health estimation for Li-ion battery using characteristic voltage intervals and genetic algorithm optimized back propagation neural network. *Journal of Energy Storage* 57 (2023), 106277.
- [15] Denzil Ferreira, Eija Ferreira, Jorge Goncalves, Vassilis Kostakos, and Anind K. Dey. 2013. Revisiting human-battery interaction with an interactive battery interface. In *Proceedings of the 2013 ACM International Joint Conference on Pervasive and Ubiquitous Computing (Zurich, Switzerland) (UbiComp '13)*. Association for Computing Machinery, New York, NY, USA, 563–572. <https://doi.org/10.1145/2493432.2493465>
- [16] Stephen Few. 2011. Quantitative displays for combining time-series and part-to-whole relationships. *Visual Business Intelligence Newsletter* (2011).
- [17] Alexandre Filipowicz, Scott Carter, Nayeli Bravo, Rumen Iliev, Shabnam Hakimi, David Ayman Shamma, Kent Lyons, Candice Hogan, and Charlene Wu. 2023. Visual Elements and Cognitive Biases Influence Interpretations of Trends in Scatter Plots. *arXiv preprint arXiv:2310.15406* (2023).

- [18] Alexandre LS Filipowicz, Nayeli Suseth Bravo, Rumen Iliev, Vikram Mohanty, Charlene C Wu, and David A Shamma. 2023. Promoting Sustainable Charging Through User Interface Interventions. In *Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. ACM Press, New York, NY, 267–279.
- [19] Steven L Franconeri, Lace M Padilla, Priti Shah, Jeffrey M Zacks, and Jessica Hullman. 2021. The science of visual data communication: What works. *Psychological Science in the public interest* 22, 3 (2021), 110–161.
- [20] Paul Gasper, Aron Saxon, Ying Shi, Elizabeth Endler, Kandler Smith, and Foram M Thakkar. 2023. Degradation and modeling of large-format commercial lithium-ion cells as a function of chemistry, design, and aging conditions. *Journal of Energy Storage* 73 (2023), 109042.
- [21] Paul Green. 1988. Human factors and gauge design: a literature review. *UMTRI Reports* (1988).
- [22] Xuebing Han, Languang Lu, Yuejiu Zheng, Xuning Feng, Zhe Li, Jianqiu Li, and Minggao Ouyang. 2019. A review on the key issues of the lithium ion battery degradation among the whole life cycle. *ETransportation* 1 (2019), 100005.
- [23] Daniel Kahneman and Shane Frederick. 2005. A model of heuristic judgment. *The Cambridge handbook of thinking and reasoning* 267 (2005), 293.
- [24] Michael L Kalish, Thomas L Griffiths, and Stephan Lewandowsky. 2007. Iterated learning: Intergenerational knowledge transmission reveals inductive biases. *Psychonomic Bulletin & Review* 14, 2 (2007), 288–294.
- [25] Ha-Kyung Kong, Zhicheng Liu, and Karrie Karahalios. 2018. Frames and slants in titles of visualizations on controversial topics. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. ACM Press, New York, NY, 1–12.
- [26] Jill H Larkin and Herbert A Simon. 1987. Why a diagram is (sometimes) worth ten thousand words. *Cognitive science* 11, 1 (1987), 65–100.
- [27] Bart de LangheStefano PuntoniRichard Larrick. 2017. Linear thinking in a non-linear world. *Harvard Business Review* "May-June 2017", 1 (2017), 130–139.
- [28] Richard P Larrick and Jack B Soll. 2008. The MPG illusion. , 1593–1594 pages.
- [29] Sharon Lin, Julie Fortuna, Chinmay Kulkarni, Maureen Stone, and Jeffrey Heer. 2013. Selecting semantically-resonant colors for data visualization. In *Computer Graphics Forum*, Vol. 32. Wiley Online Library, Wiley, Hoboken, New Jersey, 401–410.
- [30] Vikram Mohanty, Alexandre LS Filipowicz, Nayeli Suseth Bravo, Scott Carter, and David A Shamma. 2023. Save a tree or 6 kg of co2? understanding effective carbon footprint interventions for eco-friendly vehicular choices. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM Press, New York, NY, 1–24.
- [31] Tamara Munzner. 2014. *Visualization analysis and design*. CRC press, Boca Raton, FL.
- [32] California Code of Regulations. 2022. Data Standardization Requirements for 2026 and Subsequent Model Year Light-Duty Zero Emission Vehicles and Plug-in Hybrid Electric Vehicles. 13 CA ADC § 1962.5.
- [33] Stefan Palan and Christian Schitter. 2018. Prolific. ac—A subject pool for online experiments. *Journal of Behavioral and Experimental Finance* 17 (2018), 22–27.
- [34] Anshul Vikram Pandey, Anjali Manivannan, Oded Nov, Margaret Satterthwaite, and Enrico Bertini. 2014. The persuasive power of data visualization. *IEEE transactions on visualization and computer graphics* 20, 12 (2014), 2211–2220.
- [35] Dario Pevec, Jurica Babic, Arthur Carvalho, Yashar Ghiassi-Farrokhfal, Wolfgang Ketter, and Vedran Podobnik. 2020. A survey-based assessment of how existing and potential electric vehicle owners perceive range anxiety. *Journal of cleaner Production* 276 (2020), 122779.
- [36] Scott Plous. 1993. *The psychology of judgment and decision making*. Mcgraw-Hill Book Company, New York, NY.
- [37] Yuliya Preger, Heather M Barkholtz, Armando Fresquez, Daniel L Campbell, Benjamin W Juba, Jessica Romàn-Kustas, Summer R Ferreira, and Babu Chalamala. 2020. Degradation of commercial lithium-ion cells as a function of chemistry and cycling conditions. *Journal of The Electrochemical Society* 167, 12 (2020), 120532.
- [38] Ahmad Rahmati, Angela Qian, and Lin Zhong. 2007. Understanding human-battery interaction on mobile phones. In *Proceedings of the 9th International Conference on Human Computer Interaction with Mobile Devices and Services (Singapore) (MobileHCI '07)*. Association for Computing Machinery, New York, NY, USA, 265–272. <https://doi.org/10.1145/1377999.1378017>
- [39] Angela Sanguinetti, Kelsea Dombrovski, and Suhaila Sikand. 2018. Information, timing, and display: A design-behavior framework for improving the effectiveness of eco-feedback. *Energy Research & Social Science* 39 (2018), 55–68.
- [40] Julius Schmitt, Mathias Rehm, Alexander Karger, and Andreas Jossen. 2023. Capacity and degradation mode estimation for lithium-ion batteries based on partial charging curves at different current rates. *Journal of Energy Storage* 59 (2023), 106517.
- [41] Simon F Schuster, Tobias Bach, Elena Fleder, Jana Müller, Martin Brand, Gerhard Sextl, and Andreas Jossen. 2015. Nonlinear aging characteristics of lithium-ion cells under different operational conditions. *Journal of Energy Storage* 1 (2015), 44–53.
- [42] Ola Svenson. 2021. Biased judgments of the effects of speed change on travel time, fuel consumption and braking: Individual differences in the use of simplifying rules producing the same biases. *Transportation research part F: traffic psychology and behaviour* 78 (2021), 398–409.
- [43] Ola Svenson, Gabriella Eriksson, and Nichel Gonzalez. 2012. Braking from different speeds: Judgments of collision speed if a car does not stop in time. *Accident Analysis & Prevention* 45 (2012), 487–492.
- [44] Cher Ming Tan, Preetpal Singh, and Che Chen. 2020. Accurate real time on-line estimation of state-of-health and remaining useful life of Li ion batteries. *Applied Sciences* 10, 21 (2020), 7836.
- [45] Amos Tversky and Daniel Kahneman. 1981. The framing of decisions and the psychology of choice. *science* 211, 4481 (1981), 453–458.
- [46] Amos Tversky, Daniel Kahneman, and Paul Slovic. 1982. *Judgment under uncertainty: Heuristics and biases*. Cambridge, Cambridge, United Kingdom.
- [47] Patricia Valdez and Albert Mehrabian. 1994. Effects of color on emotions. *Journal of experimental psychology: General* 123, 4 (1994), 394.
- [48] Zuolu Wang, Guojin Feng, Dong Zhen, Fengshou Gu, and Andrew Ball. 2021. A review on online state of charge and state of health estimation for lithium-ion batteries in electric vehicles. *Energy Reports* 7 (2021), 5141–5161.
- [49] Zengkai Wang, Shengkui Zeng, and Jianbin Guo. 2021. Understanding the influence of state of health on the range anxiety of battery electric vehicle drivers. *IET Intelligent Transport Systems* 15, 2 (2021), 286–296.
- [50] Lei Wu, Lingli Guo, Hao Fang, and Lijun Mou. 2019. Bullet graph versus gauges graph: Evaluation human information processing of industrial visualization based on eye-tracking methods. In *Advances in Usability, User Experience and Assistive Technology: Proceedings of the AHFE 2018 International Conferences on Usability & User Experience and Human Factors and Assistive Technology, Held on July 21–25, 2018, in Loews Sapphire Falls Resort at Universal Studios, Orlando, Florida, USA 9*. Springer, 752–762.
- [51] Cindy Xiong, Joel Shapiro, Jessica Hullman, and Steven Franconeri. 2019. Illusion of causality in visualized data. *IEEE transactions on visualization and computer graphics* 26, 1 (2019), 853–862.
- [52] Fumeng Yang, Mandi Cai, Chloe Mortenson, Hoda Fakhari, Ayse D Lokmanoglu, Jessica Hullman, Steven Franconeri, Nicholas Diakopoulos, Erik C Nisbet, and Matthew Kay. 2023. Swaying the Public? Impacts of Election Forecast Visualizations on Emotion, Trust, and Intention in the 2022 US Midterms. *IEEE Transactions on Visualization and Computer Graphics* 30, 1 (2023), 23 – 33.